

Graduation Project proposal

Higher Technological Institute
Computer Science Department

Supervisor(s):

- | | |
|----------------------|---------------------------|
| 1- DR. Mona Mohammed | 2- DR. Farahat |
| 3- DR. Sara Ahmed | 4- DR. Magdy abd el ghani |

Student name and ID: (maximum 6 students)

Student ID	Student Name
42020285	Moaaz Ahmed Abdulfattah Mohammed
42021352	Mahmoud Attia Fathy Elsaid
42021217	Mustafa Hany Mustafa Abdulhalim
42021348	Abdulrahman Atif Abdullah Hussein
42021324	Mahmoud Mohammed Elhusseini Mohammed
42021358	Anas Elsaid Abdulmaati Ahmed

Project title (or topic):

FastAF

Short project description:

This graduation project aims to revolutionize the traditional cashier and database systems in pharmacies by developing an innovative solution that leverages the power of deep learning and computer vision technologies. The primary goal is to develop a highly efficient and user-friendly solution that automates and streamlines pharmacy operations.

The project will develop an Android application capable of efficiently scanning and recognizing pharmaceutical products using the device's camera. Upon scanning, the system will accurately identify the drug, retrieve its relevant information, and automatically update the central database accordingly through the web app, ensuring real-time up-to-date inventory records and sales data. This will eliminate the need for manual barcode scanning, reducing errors and speeding up the checkout process. The web application will have a user-friendly interface that features:

Daily Sales Monitoring: record and display daily sales and generate sales reports enabling pharmacists for easy transaction monitoring and assess the financial performance of the pharmacy.

Inventory Management and Alerts: The system will keep an updated record of the inventory and provide alerts for drugs that are running low. Additionally, it will monitor expiry dates and notify users about products nearing expiration to reduce wastage and ensure patient safety.

Data Analytics Dashboard: Offers valuable insights into sales patterns and customer behavior through visualized data on sales trends and inventory dynamics to aid in making data-driven decisions to optimize stock and improve operational efficiency.

The proposed project not only aims to enhance the operational efficiency of the cashier process in pharmacies but also seeks to minimize human error, save time, improve customer service, ensure accurate inventory management and timely identification of expired drugs, and streamline the overall management process. By harnessing the power of deep learning and computer vision, this system aspires to set a new standard for modern pharmacy management, ultimately contributing to better productivity and healthcare outcomes.

Technical assignment:

Backend (language : java, framework : Spring Boot, DB : PostgreSQL MongoDB Redis Elastic) | **Frontend** (Language : JavaScript, Framework : React, axios) | **UI/UX** (Framework : figma) | **App** (Java/kotlin for Android developing, Tools : Android studio code) | **AI** (language : python, framework : tensorflow/keras/scikit-learn, yolo, OpenCV, numpy) | **Version control** (git and GitHub)

Department Head

()

Dr/Yousrya Abo el naga